Paper Information

Title: A Survey on Machine Learning for Recurring Concept Drifting Data Streams. **Authors:** Andrés L. Suárez-Cetrulo, David Quintana, and Alejandro Cervantes. **Journal:** *Expert Systems with Applications*, volume 213, 2023, article 118934. **DOI:** 10.1016/j.eswa.2022.118934.

Research Problem and Objectives

Data streams arrive continuously, and their patterns can change over time. These changes, called concept drift, can make older models inaccurate. Some changes later repeat, as with seasonal patterns. Standard methods may forget earlier patterns and need to relearn them, costing time and computing resources. The authors review ways to detect changes and reuse earlier models as new data arrives, focusing on recurring patterns rather than repeated batch retraining.

Methodology and Proposed Approach

This literature survey organizes earlier work into supervised online ensembles, meta-learning and model reuse, and unsupervised or model-based clustering. It reviews passive methods, which update continuously, and active methods, which use drift detectors to trigger a response. Some systems retain past models and reuse them when a familiar pattern returns; others represent concepts as clusters to detect change beyond prediction error alone. The authors compare design choices and reported evaluations on data streams, including synthetic datasets. These studies use measures such as accuracy, Kappa, runtime, and memory. The survey itself runs no new experiment and defines no new dataset or baseline.

Key Findings and Contributions

Ensembles and meta-learning offer ways to reuse models when concepts recur, potentially reducing relearning and improving predictions. The survey reports findings from earlier studies, not new experiments. Adaptive Random Forest (ARF) often performs strongly in classification comparisons, although other methods lead on some streams. Clustering can complement supervised drift detection, especially when accuracy alone is misleading. For real-time analysis, the paper's main contribution is an overview of methods that balance adaptation, model reuse, and computing limits.

Limitations and Future Research

Many surveyed methods focus on classification, while regression and unsupervised settings need better drift-handling methods. Real-world evaluation is difficult when true change points are unknown. Storing past models can also require substantial memory and computation. As a survey, the paper does not directly compare all methods in one controlled experiment, so results from separate studies may not be comparable. The authors suggest methods for regression, better ways to simulate or label realistic drift, and strategies to remove unused models. They also call for stronger links to time-series research, transfer learning, and deep learning.

Relevance to Our Research Direction

This paper directly relates to real-time analysis: it examines learning from changing data without stopping for full retraining. Our group could explore model reuse, concept histories, or clustering-based drift detection. One research question is whether a system can recognize a returning pattern and reuse its model within response-time and memory limits. Testing on real streams, where change points may be unknown, is another direction. The survey also highlights gaps beyond classification, including regression.

Paper Information

Title: A Benchmark and Survey of Fully Unsupervised Concept Drift Detectors on Real-World Data Streams. **Authors:** Daniel Lukats, Oliver Zielinski, Axel Hahn, and Frederic Stahl. **Journal:** *International Journal of Data Science and Analytics*, volume 19, 2025, pages 1–31. **DOI:** 10.1007/s41060-024-00620-y.

Research Problem and Objectives

Data streams continuously generate new information, and their patterns may change over time. This is called concept drift. When such changes occur, machine learning models trained on older data may become less accurate. Many existing drift detection methods require correct class labels to identify these changes. However, in real-world applications, obtaining labels immediately can be expensive or impossible. The authors aim to review and compare fully unsupervised concept drift detectors that can identify changes without requiring labeled data. They also investigate which detectors perform well and whether commonly used evaluation metrics provide reliable results.

Methodology and Proposed Approach

The authors searched Google Scholar and Semantic Scholar for existing methods and selected 10 fully unsupervised concept drift detection algorithms that met their inclusion criteria. They examined each algorithm's main idea, architecture, data requirements, and assumptions. Seven algorithms were implemented and evaluated on 11 real-world data streams, including insect sensor data, electricity usage, weather, and other datasets. The evaluation used different measures, including classifier accuracy, lift-per-drift (LPD), and Mean Time Ratio (MTR). Four online classifiers were also used to examine how drift detection affects predictive performance. The remaining three algorithms were excluded from the experiments because their implementations were too slow or depended heavily on randomness. The implementations, configurations, and experimental results were made publicly available to support reproducibility.

Key Findings and Contributions

The experiments show that no single detector performs best under every evaluation metric. The Image-Based Drift Detector (IBDD) achieves strong classifier accuracy on many data streams, while the Discriminative Drift Detector (D3) and Semi-Parametric Log-Likelihood (SPLL) perform well under multiple evaluation measures. The authors recommend choosing a detector according to the application's priorities. The study also finds that classifier accuracy can favor detectors that trigger more frequent model adaptations, while LPD can favor detectors that trigger fewer detections. Therefore, these metrics may not always reflect the actual quality of drift detection. The main contributions are a systematic survey, seven implemented detectors, a benchmark using 11 real-world streams, and an analysis of weaknesses in existing evaluation metrics.

Limitations and Future Research

Exact information about when concept drift occurs is unavailable for most real-world data streams, making direct evaluation difficult. MTR could only be calculated for one data stream, limiting comparisons using this metric. Some algorithms were too slow for thorough online evaluation, while others depended heavily on randomness. The study also identifies problems with existing metrics, particularly their tendency to favor certain detection behaviors.

Although the study evaluates several detectors on real-world data, its results do not establish which detector will perform best in every application. Performance may depend on the data characteristics, parameter settings, and evaluation metric. Further testing on additional streams would help determine how well the findings generalize.

The authors identify the need for better evaluation methods for gradual and incremental drift, improved ways to choose data-window sizes, and the use of AutoML to help configure detectors. They also suggest improving computational efficiency, integrating the best-performing detectors into online learning frameworks, and combining different detectors into a heterogeneous ensemble.

Relevance to Our Research Direction

This paper is directly relevant to real-time streaming analysis because it investigates how to detect changing patterns in continuously arriving data without relying on immediate access to labels. Our group could explore unsupervised drift detection for applications such as sensor monitoring, network traffic analysis, or real-time business data. One possible research question is whether an adaptive detector can identify changes quickly while minimizing false alarms and computational costs. Another direction is to compare multiple detectors on the same streaming dataset and investigate

whether combining them produces more reliable results. The paper also highlights the importance of selecting suitable evaluation metrics, especially when the true drift points are unknown.

Paper Information

Title: FAC-fed: Federated adaptation for fairness and concept drift aware stream classification. **Authors:** Maryam Badar, Wolfgang Nejdl, and Marco Fisichella. **Journal:** *Machine Learning*, volume 112, 2023, pages 2761–2786. **DOI:** 10.1007/s10994-023-06360-7.

Research Problem and Objectives

Real-time data streams continuously generate new information, and their patterns may change over time. This change is known as concept drift, and it can reduce the predictive performance of machine learning models. Another important problem is discrimination, where a model produces unfair predictions for different groups of people. These problems become more challenging in federated learning because data is stored on separate clients and cannot always be collected on a central server due to privacy concerns. Previous studies have mainly addressed fairness or concept drift separately, with limited attention to both problems in federated stream learning. The authors aim to develop a method that reduces discrimination, handles concept drift, and maintains good predictive performance without requiring clients to share their private data.

Methodology and Proposed Approach

The authors propose FAC-Fed, a federated learning framework designed for fairness-aware and concept drift-aware stream classification. Each client maintains its own data stream and an online deep neural network (ODNN). The clients train their local models using newly arriving data and share model parameters with a central server, which averages these parameters to update the global model. The framework uses the Early Drift Detection Method (EDDM) to identify concept drift and update the sliding window of recent data. It measures discrimination using two fairness metrics: statistical parity (StP) and equal opportunity (EqOp). When discrimination exceeds a predefined threshold, the Continuous Fairness-aware Synthetic Oversampling Technique (CFSOTE) generates additional samples from selected groups in the sliding window to reduce unfairness. The authors evaluate FAC-Fed on four datasets: Bank Marketing, Law School, Default, and Adult Census. They compare its centralized version with existing methods, including FABBOO, FAHT, and CSMOTE. Balanced accuracy and G-mean are used to measure predictive performance, while StP and EqOp measure fairness.

Key Findings and Contributions

The experimental results show that FAC-Fed can reduce discrimination while maintaining good predictive performance. It achieves low discrimination scores under both statistical parity and equal opportunity, demonstrating that the framework can work with different fairness measures. The method also performs consistently when data is distributed unevenly across clients, known as non-independent and identically distributed (non-IID) data. Compared with existing centralized baseline methods, FAC-Fed generally achieves better fairness and competitive or higher predictive performance. Its fairness-aware mechanism also has only a limited impact on predictive utility compared with a version that does not address fairness. The main contributions are a federated framework that handles fairness and concept drift together, a new adaptive data augmentation technique called CFSOTE, and an experimental evaluation on multiple real-world datasets with different data distributions.

Limitations and Future Research

The authors explain that fairness-aware federated baselines for streaming data were not available for direct comparison. Therefore, they compared the centralized version of FAC-Fed with existing centralized methods rather than conducting a direct comparison with other fairness-aware federated stream-learning frameworks. The experiments use four datasets, and most of them are static datasets adapted to a federated setting. The authors also identify asynchronous training with a large number of clients and continuously arriving data streams as an area for further development.

Although FAC-Fed demonstrates promising results, its evaluation on a limited number of datasets leaves questions about how well it generalizes to other real-time applications. The framework also depends on design choices such as the drift detection method, sliding-window behavior, fairness threshold, and oversampling parameters. Testing these choices in more complex streaming environments would help determine the method's practical scalability and reliability.

The authors specifically suggest extending FAC-Fed to support asynchronous training with a large number of clients receiving continuous data streams. Further research could also evaluate the framework on additional real-world streaming

datasets and investigate its communication overhead, computational cost, and behavior under different types of concept drift.

Relevance to Our Research Direction

This paper is highly relevant to real-time streaming analysis because it addresses changing data patterns, distributed data sources, and the need to maintain reliable predictions as new data arrives. It demonstrates how a streaming system can adapt its model while also considering fairness and data privacy. Our group could explore combining concept drift detection with adaptive model updates in applications such as financial transaction monitoring, healthcare data analysis, or network traffic monitoring. Another possible direction is to investigate how different drift detection methods affect predictive accuracy, fairness, and processing time in a distributed streaming environment. We could also study whether asynchronous federated learning can improve scalability when many clients continuously generate data.

Paper Information

Title: One or Two Things We Know About Concept Drift—A Survey on Monitoring in Evolving Environments. Part B: Locating and Explaining Concept Drift

Authors: Fabian Hinder, Valerie Vaquet, and Barbara Hammer

Journal: Frontiers in Artificial Intelligence, Volume 7 (2024)

DOI: 10.3389/frai.2024.1330258

Research Problem and Objectives

Concept drift occurs when data patterns or distributions change over time, which can reduce the accuracy of machine learning systems. Although many studies focus on detecting when drift occurs, detection alone does not explain which parts of the data have changed or what makes the change important. This paper reviews existing methods for locating and explaining concept drift, compares selected localization techniques, and identifies challenges that need further research.

Methodology and Proposed Approach

The authors review existing techniques and compare three drift localization methods: kdq-Trees, LDD-DIS, and Model-Based Drift Localization (MB-DL). They conduct experiments using synthetic data to examine how sample size, drift strength, data dimensionality, and feature correlations affect localization performance. The methods are evaluated using ROC-AUC, which measures how well they distinguish affected data points from unaffected ones. The paper also discusses approaches for explaining drift in a way that people can understand.

Key Findings and Contributions

The experimental results show that MB-DL performs best overall, followed by LDD-DIS and kdq-Trees. Larger sample sizes and stronger drift generally make changes easier to locate, while high-dimensional data can reduce performance. Tree-based methods may struggle when drift involves relationships between features. The paper provides a structured overview of drift localization and explanation and highlights model-based explanations as a promising way to understand what changes in evolving data.

Limitations and Future Research

The experimental comparison is limited to a synthetic dataset, so the results may not represent every real-world streaming environment. Localization also becomes difficult when data has many features or complex relationships. Future research should evaluate methods on diverse real-world datasets, improve scalability for high-dimensional data, and develop clearer explanations. User studies could also help determine whether these explanations are useful for people monitoring real-time data streams.

Relevance to Our Research Direction

This paper is relevant to *Real-Time Streaming Analysis for Big Data* because data patterns can change continuously in streaming environments. It shows that effective monitoring should do more than detect drift: it should also identify where changes occur and explain them. These ideas can support research into scalable streaming systems that recognize changing patterns and provide understandable information for further analysis and decision-making.

Paper Information

Title: Multi-type Concept Drift Detection under a Dual-layer Variable Sliding Window in Frequent Pattern Mining with Cloud Computing. **Authors:** Jing Chen et al. **Journal:** *Journal of Cloud Computing*, Volume 13, Article 40, 2024. **DOI:** 10.1186/s13677-023-00566-9.

Research Problem and Objectives

Real-time data streams continuously generate large amounts of data whose patterns may change over time. Fixed-size sliding windows may fail to adapt efficiently to these changes, affecting frequent pattern mining. The authors aim to develop a method that detects different types of concept drift while dynamically adjusting the window size to improve the accuracy and efficiency of mining evolving data streams.

Methodology and Proposed Approach

The authors propose DLVSW-CDTD, a dual-layer variable sliding window algorithm for concept drift detection in frequent pattern mining. It uses short and long windows to identify abrupt and gradual drift, respectively. The window size changes according to detected changes in data distributions and frequent patterns. The method also applies time-decay weights to reduce the influence of outdated data. Experiments use synthetic datasets and the BMS-Webview-1 dataset to evaluate drift detection, window adjustment, execution time, and memory consumption.

Key Findings and Contributions

The results show that the proposed algorithm can detect multiple types of concept drift and adjust its window size accordingly. Compared with fixed-window approaches, it better adapts to changing data patterns while maintaining stable performance. The authors also report efficient processing and memory usage. Its main contribution is combining adaptive window sizing, multi-type drift detection, and drift adaptation within a frequent pattern mining framework for cloud-based data streams.

Limitations and Future Research

The evaluation focuses mainly on frequent pattern mining and includes generated datasets, limiting conclusions about other real-world streaming tasks. The algorithm also requires initial parameter and threshold settings. The authors suggest further optimization for cloud environments, including distributed computing, parallelization, and improved resource allocation to enhance scalability.

Relevance to Our Research Direction

This paper is directly relevant to adaptive window selection for real-time Big Data streams. It demonstrates how changing window sizes can help detect concept drift while balancing recent and historical information. Our research could investigate how adaptive window strategies perform across different drift types and datasets, measuring detection delay, false alarms, processing time, and memory usage, particularly for unsupervised drift detection.

Paper Information

Title: A Novel Edge Architecture and Solution for Detecting Concept Drift in Smart Environments. **Authors:** Hassan Mehmood et al. **Journal:** *Future Generation Computer Systems*, Volume 150, 2024, pp. 127–143. **DOI:** 10.1016/j.future.2023.08.023.

Research Problem and Objectives

Data streams from smart buildings and cities continuously change due to factors such as sensor malfunctions, environmental variations, and changes in human activities. These changes, known as concept drift, can reduce the predictive accuracy of machine learning models. The authors aim to develop an edge-computing architecture that detects concept drift and adapts models while supporting real-time data processing with low latency.

Methodology and Proposed Approach

The authors propose an Edge Micro Data Centre (EMDC) architecture that combines edge and cloud resources using containerisation and workload migration. For drift detection, they integrate a Long Short-Term Memory (LSTM) model with three methods: Page–Hinkley Test (PHT), Adaptive Windowing (ADWIN), and Kolmogorov–Smirnov Windowing (KSWIN). The approach is evaluated using synthetic datasets with known drift points and real-world CO_2 sensor data collected from five locations at the University of Oulu Smart Campus. Model performance is measured using MAE, RMSE, and MAPE.

Key Findings and Contributions

The results demonstrate that integrating drift detection improves prediction accuracy compared with using LSTM alone. KSWIN achieves the best performance, reducing MAPE from 8.5% to 3.88%. The main contributions are the EMDC architecture, automated feedback-based drift detection and model adaptation, and support for migrating computational workloads between edge and cloud environments.

Limitations and Future Research

The evaluation focuses on a specific smart-campus CO_2 forecasting use case, limiting the generalisability of the findings. The approach may require manual threshold tuning, and further testing is needed to assess false-positive detections. The authors suggest evaluating additional smart-city applications and using larger datasets to improve model tuning and reliability.

Relevance to Our Research Direction

This paper is relevant because it compares ADWIN with other drift detectors in real-time sensor streams. However, its primary focus is edge architecture and prediction adaptation rather than adaptive window selection itself. Our research could build on this work by comparing window-selection strategies across different drift types and datasets, evaluating detection delay, false alarms, processing time, and memory usage.

Paper Information

Title: One or Two Things We Know About Concept Drift—A Survey on Monitoring in Evolving Environments. Part A: Detecting Concept Drift. **Authors:** Fabian Hinder, Valerie Vaquet, and Barbara Hammer. **Journal:** *Frontiers in Artificial Intelligence*, Volume 7, 2024. **DOI:** 10.3389/frai.2024.1330257.

Research Problem and Objectives

Concept drift occurs when the underlying data distribution changes over time. Such changes can cause monitoring systems to miss anomalies or behave incorrectly. While existing surveys often focus on supervised learning, this paper addresses unsupervised drift detection, where ground-truth labels are not required. Its objectives are to organize existing detection methods, formally define the detection problem, and compare different approaches systematically.

Methodology and Proposed Approach

The authors conduct a literature survey and develop a taxonomy of unsupervised drift detection methods. They describe a general four-stage detection process: selecting data windows, building data descriptors, calculating dissimilarity, and normalizing the resulting scores. The survey categorizes methods into two-sample analysis, meta-statistic-based, and block-based approaches. It also presents standardized experiments on artificial datasets to compare detection strategies and examine their suitability under different streaming conditions.

Key Findings and Contributions

The paper provides a structured overview of unsupervised drift detection and explains the strengths and limitations of different methods. It discusses techniques such as the Kolmogorov–Smirnov test, Maximum Mean Discrepancy (MMD), and Adaptive Windowing (ADWIN). The authors highlight the importance of window selection, statistical validity, computational cost, and false-alarm control. They also explain that detector performance depends on the characteristics of the data stream and the type of drift.

Limitations and Future Research

The experimental comparisons use parametric artificial datasets, so their results may not fully represent complex real-world streams. Detection performance also depends on factors such as sample size, drift strength, window selection, and the chosen statistical measure. Further evaluation on diverse real-world data streams is important for understanding practical performance and scalability.

Relevance to Our Research Direction

This paper is highly relevant to our research on adaptive window selection for unsupervised concept drift detection. It explains how data-window selection forms the first stage of many detection methods and influences subsequent comparisons between data distributions. Our research can build on this foundation by experimentally comparing fixed and adaptive window strategies under different drift types, measuring detection delay, false alarms, processing time, and memory usage.

Paper Information

Title: Scalable Concept Drift Adaptation for Stream Data Mining. **Authors:** Lisha Hu, Wenxiu Li, Yaru Lu, and Chunyu Hu. **Journal:** *Complex & Intelligent Systems*, Volume 10, pp. 6725–6743, 2024. **DOI:** 10.1007/s40747-024-01524-x.

Research Problem and Objectives

Concept drift occurs when the underlying data distribution changes over time, reducing the effectiveness of existing predictive models. Traditional offline learning algorithms struggle to adapt quickly to continuously arriving data. Existing single-pass methods based on Minimum Enclosing Ball (MEB) mainly expand the ball to include new data, potentially making it excessively large and preventing effective model updates. This paper aims to develop a scalable online learning framework that adapts to changing concepts while processing each sample only once.

Methodology and Proposed Approach

The authors propose Scalable Concept Drift Adaptation (SCDA), a framework based on Support Vector Machines (SVM) and Minimum Enclosing Ball (MEB). It introduces three methods: SCDA-I, SCDA-II, and SCDA-III. Unlike approaches that only expand the ball, these methods dynamically expand, contract, or reposition it according to incoming samples. The methods are evaluated on seven synthetic datasets and five real-world datasets, covering gradual, sudden, and incremental drift. Comparisons are conducted against StreamSVM, OSVM-SP, Naive Bayes, Hoeffding Tree, and ADWIN.

Key Findings and Contributions

The experimental results show that SCDA-III achieves the best overall performance, with the lowest average prequential error rank of 1.33 across the eight compared methods. It also demonstrates effective memory usage and better control of the enclosing ball's size. The results indicate that allowing the model to contract or reposition its ball can improve adaptation compared with methods that only expand it. However, individual methods perform better on some real-world datasets.

Limitations and Future Research

The study focuses exclusively on binary classification and uses balanced synthetic datasets. The authors identify multiclass classification, imbalanced data, and selecting a small number of informative samples from massive streams as future research directions. Furthermore, the drift types of the real-world datasets are unknown, limiting direct evaluation of detection timing on those datasets.

Relevance to Our Research Direction

This paper provides a useful baseline for understanding how online models adapt to concept drift. Its discussion of ADWIN also connects to adaptive windowing, where the window size changes in response to data changes. However, SCDA focuses on dynamically adjusting a model's enclosing ball rather than selecting a data-window size. Our research can investigate adaptive window selection directly and compare its detection delay, false-alarm rate, processing time, and memory usage under different drift types.